

Workshop on Trust and Reliance in AI-Human Teams (TRAIT)

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ABSTRACT

As humans increasingly interact (and even collaborate) with AI systems during decision-making, creative exercises, and other tasks, *appropriate* trust and reliance are necessary to ensure proper usage and adoption of these systems. Specifically, people should understand when to trust or rely on an algorithm's outputs and when to override them. Significant research focus has aimed to define and measure trust in human-AI interaction, and design and implement interactions that promote and calibrate trust. However, conceptualizing trust and reliance, and identifying the best ways to measure these constructs and effectively shape them in human-AI interactions remains a challenge, especially across contexts and domains.

This workshop aims to establish building appropriate trust and reliance on (imperfect) AI systems as a vital, yet under-explored research problem. The workshop will provide a venue for exploring three broad aspects related to human-AI trust: (1) How do we clarify definitions and frameworks relevant to human-AI trust and reliance (e.g., what does trust mean in different contexts)? (2) How do we *measure* trust and reliance? And, (3) How do we *shape* trust and reliance? The workshop will build on the success from running it at CHI 2022,¹ with a focus on "Learning from Practice" — how can we better tie theory-building to real-life use cases? As these problems and solutions involving humans and AI are interdisciplinary in nature, we invite participants with expertise in HCI, AI, ML, psychology, and social science, or other relevant fields to foster closer communications and collaboration between multiple communities.

*Equal contribution from all authors. Names ordered alphabetically.

¹<https://chi-trait.github.io/>

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CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *Interactive systems and tools*; • **Computing methodologies** → Machine learning.

KEYWORDS

trust, human-centered artificial intelligence, reliance, uncertainty

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1 INTRODUCTION AND BACKGROUND

AI is increasingly deployed in people-facing systems to assist and collaborate with users (e.g., domain experts) through their outputs, such as for assisted decision making and creative tasks. Consequently, interest in human-AI interaction and collaboration is exploding both within and beyond the HCI community [1, 9, 22, 24]. But when is a human-AI collaboration “successful”—and how do we measure success in these contexts—and what factors (of people, AI, interface, task, etc) impact the success of the human-AI teams? One set of frequently discussed, relevant constructs are human-AI “trust” and “reliance.” For example, many studies have started to investigate the impact that system transparency and accuracy have on trust [4, 7, 11, 25]; the role of trust in the user experience of complex algorithmic systems, such as the impact on satisfaction, adoption, and reliance [5, 15]; and the potentially undesirable effects that too much or too little trust might have on a user's (over- or under-) reliance upon a given system [2, 13, 19, 20]. Despite all this interest, the community still lacks a conceptual understanding of definitions of human-AI trust and reliance in the context of successful human-AI collaborations [5, 8, 10].

In practice, to ensure that human-AI collaborations do more good than harm, it is vital that we understand, measure, and shape human-AI trust and reliance; for example, when is user trust and

reliance on AI warranted or *appropriate*, how does it evolve over the course of (long-term) human-AI collaborations, which factors (individual, contextual, etc) influence it, and how can we design interventions that guard against harms caused by inappropriate trust or reliance and instead promote more appropriate reliance within human-AI collaborations? Indeed, recent years have seen renewed interest in human-AI trust within the AI and HCI community [12, 16, 18, 21, 23] motivated by directions and approaches introduced in prior literature on human-automation interaction and human factors [14]. However, it is not yet well understood what generalizes from these works and what new considerations modern human-AI collaborations introduce [3, 8].

Conceptualizing trust and reliance, formalizing methods to measure them, and studying and shaping their impact on human-AI performance remains challenging for many reasons. For example, defining and measuring human “trust” in an AI system is subject to questions that have long plagued other behavioral sciences, such as the extent to which researchers should focus on *realized behaviors* when interacting with an AI system (e.g., adoption, reliance, or adherence) versus the *beliefs* that give rise to those behaviors [5, 8]. Shaping human-AI trust and reliance may be even cumbersome—Even in the context of simple, ubiquitous human-AI teams, such as AI-assisted decision making, trust may depend on many aspects of the human’s mental model: how they conceive the task, their domain expertise, how they conceive of the AI’s objectives and decision strategies, their understanding of the AI’s capabilities and limitations, and (their understanding of) their own capabilities and limitations, to name just a few. While (post-deployment) AI systems are often conceived as independent of its developers, in real-world organizational settings, system trust may in fact depend on the user’s trust in those developers [6], and humans may choose to rely on an AI system for reasons unrelated to their trust in those systems (e.g., organizational pressures or incentives). Finally, “trust” is relevant to human-AI collaboration in large part because such collaborations often involve decision-making under uncertainty. And, behavior under uncertainty is complex and humans decisions under uncertainty exhibit many biases that depend heavily on context. For example, people can be either under- and over-sensitive to important features of a decision task like sample size or variance, depending on the specific situation. As a result, recent results indicating under- or over-reliance in human-AI collaboration may be more context-dependent than is commonly appreciated.

As research in the areas of human-AI interaction and collaboration grows, it is increasingly important to synthesize empirical results across contexts and conceptualize and formalize key concepts. In the absence of a theoretical basis that grounds our understanding, findings from research (e.g., focused on socio-technical approaches) on improving trust and reliance in human-AI collaborations can seem ad-hoc and at times contradictory. For example, prior research has found that exposing system confidence (or uncertainty) can both help [26] or hinder [17] trust calibration. Without a synthesized understanding of research space around trust, reliance and the effectiveness of human-AI collaboration, the risk is that this body of research will continue to expand in size, but not necessarily in insight and impact on the real-world.

1.1 Workshop Goals

Our goal is to **establish appropriate trust and reliance on (imperfect, people-facing) AI systems as a vital yet under-explored research problem**, especially for developing human-AI interactions that effectively *augment* users. We wish to highlight incentives and challenges from pursuing this problem and help spur impactful future research. This year, our workshop builds on its previous success with a focus on “Learning from Practice” — how can we better tie theory-building to real-life use cases?

We aim to **synthesize (and conceptualize) various definitions and frameworks relevant to human-AI trust and reliance**. Many prior works across research sub-communities (such as, HCI, AI, ML, NLP, psychology, etc) motivate their research using concepts and terms similar to “trust” and “reliance” in AI; however, we notice a lack of consensus on many dimensions: (1) What does “trust” mean in different contexts? (2) When should users trust AI and when should trust in AI be defined as warranted or appropriate? (3) What potential harms would inappropriate trust cause in different domains? Thus through this workshop, we wish to synthesize domain-agnostic or -specific themes, definitions, known observations and results, and key challenges to inform future research. We plan to include a broad perspective on human-AI trust by including topics on (and discussions around) (1) people interacting with complex algorithmic systems (e.g., in various domains and social contexts, with varying levels of prior experiences and biases), (2) human-human interaction and group dynamics, and (3) trust from the perspective of machine learning systems (e.g., calibrating AI’s confidence, explainable AI, etc).

In accordance with these key themes and challenges, we also would like spur research on *measuring* and *shaping* trust and reliance. **Measuring trust and reliance** is a pre-requisite for developing mechanisms that lead to appropriate trust and reliance. Such research may include categorizing objective and subjective trust (e.g., through quantitative human-AI team performance and qualitative user reflection) and eliciting humans’ mental models on AI systems, as well as identifying factors (e.g., system uncertainty, mental model, decision risk, etc.) that affect humans’ trust and reliance. We also aim to further **promote research on developing mechanisms to shape trust and reliance with respect to the factors**, e.g., by calibrating user expectations on model capabilities through careful user training and on-boarding, by designing human-AI team architectures, visualizations and interfaces that supports more effective human-AI communications in a team setting, etc.

To summarize, the workshop will focus on three broad aspects:

- (1) How do we *clarify* definitions and frameworks relevant to human-AI trust and reliance?
- (2) How do we *measure* trust and reliance and identify factors that affect these constructs?
- (3) How do we *shape* trust and reliance toward effective human-AI collaboration?

As the problems and solutions involving AI and people are interdisciplinary in nature, we will invite people with expertise in HCI, AI, ML, psychology, social sciences, or other relevant fields to foster closer communication and collaborations between multiple communities.

1.2 Theme at CHI 2023: “Learning from Practice”

Last year, we organized the TRAIT workshop at CHI 2022, which was well attended. Our workshop received nearly 60 submissions, and we selected 42 submissions for presentation (with ten spotlight, long presentations); more than 90 researchers joined in a hybrid setting from eight time zones. It is evident from the overwhelming number of participants that the topic is of great interest.

Due to the novelty of theories and measures of trust in the CHI community, last year’s workshop focused on defining trust and surfacing methods and theories across disciplinary boundaries. The 2022 workshop collected a series of theoretical framing and evaluations through keynote talks, paper presentations, and group activities.² We are currently in the progress of writing a blog post and a manuscript summarizing our findings. However, for HCI researchers’ contributions to the topic of appropriate trust in human-AI teams to be applicable in real world settings, it is important that their contributions informed by real-world use cases, constraints, data availability, and cost-benefit trade-offs. Without explicit attempts to incorporate examples from practice, it remains unclear how useful trust frameworks and methodologies will be in complex real-world settings. Building on TRAIT version one, we plan run the workshop again at CHI 2023 with an emphasis on **Learning from Practice**. The workshop will still welcome contributions related to the many dimensions of trust covered last year, but integrate additional emphasis on understanding industrial and other organizational perspectives through an invited keynote, panel, and activities. Example questions that will motivate this year’s workshop include: How concerned are practitioners about trust and reliance? What are motivations in non-academic settings for measuring or otherwise assessing trust? And how do their motivations differ from academia’s? What potential role, if any, could emerging theories of appropriate trust in HCI play in these situations? How does the nature of available data in organizations shape how trust is conceived? We will also adjust our workshop design to make it more welcoming to industry perspectives (e.g., adding an industry track to our Call for Papers). In doing so, we will carry over last year’s objective of being inclusive. Additionally, we address some participant feedback from last year.

2 ORGANIZERS

We will re-run TRAIT with the same organization team as last year (ordered alphabetically).

Gagan Bansal is a Senior Researcher at Microsoft Research, Redmond. He received his Ph.D. degree from the Allen School of Computer Science and Engineering at the University of Washington, Seattle. He conducts interdisciplinary research on Artificial Intelligence and Human-Computer Interaction with focus on developing human-centered AI systems for augmenting people.

Zana Bućinca is a Ph.D. Candidate at Harvard University. Her research lies at the intersection of Human-Computer Interaction and Artificial Intelligence. Informed by cognitive science theories, Zana designs, builds, and evaluates AI for decision-making support.

Kenneth Holstein is an Assistant Professor in Human-Computer Interaction at Carnegie Mellon University. His research interests lie at the intersection of HCI, AI, design, and cognitive science, focusing on the design, development, and evaluation of human-AI collaborative systems in complex social contexts.

Jessica Hullman is an Associate Professor of Computer Science at Northwestern University. Her research looks at how to design, evaluate, coordinate, and think about visual representations of data and model predictions for inference, decision making, and communication, including the effects of visualizing uncertainty on belief updating and potential for behaviorally induced feedback loops in visualizing model predictions in strategic settings.

Alison Smith-Renner is a Senior Research Scientist at Dataminr. Her research interests lie at the intersection of AI and HCI, focusing on transparency and control for human-in-the-loop systems to engender appropriate trust and improve human performance. Alison received her Ph.D. from the University of Maryland, College Park. She has organized various workshops on explainable AI and human-centered ML, including at IUI, CHI, and TEI, and she has held senior committee roles at IUI.

Simone Stumpf is a Reader in Responsible and Interactive AI at University of Glasgow, UK. She has a long-standing research focus on user interactions with machine learning systems. Her work has contributed to shaping the field of Explainable AI (XAI) through the Explanatory Debugging approach to interactive machine learning, providing design principles for crafting explanations. She has been a member of the organising committee of the ExSS workshop at IUI, and has held senior committee roles at CHI, IUI and EICS conferences.

Sherry Tongshuang Wu is an Assistant Professor in the Human-Computer Interaction Institute at Carnegie Mellon University. Her research lies at the intersection of Human-Computer Interaction and Natural Language Processing, aiming to design, evaluate, build, and interact with AI systems that are compatible with actual human goals. Before joining CMU, Sherry received her Ph.D. degree from the University of Washington.

3 WEBSITE

We will reuse and update our website at <https://chi-trait.github.io> to advertise and disseminate the workshop’s information and call for proposals. We will also use this website to share workshop contributions, including accepted papers, and support future engagement.

4 PRE-WORKSHOP PLAN

In addition to the website, we will advertise the workshop through email distribution lists at relevant conferences and research institutions (including, but not limited to, FAccT, IUI, CHI, ACL, and CSCW mailing lists), direct communication with colleagues in the field, and social media.

We will have a program committee (PC) with experts from diverse research organizations and backgrounds who will help us to curate the workshop by disseminating the call for papers and reviewing submissions. Last year, we had commitments from 33 PC members with expertise in human-AI interaction and related

²See one sample jamboard screenshot here: https://twitter.com/bansalg/status/1520491084643721221?s=20&t=Z4hZE_hVhWfB4U3ZQP4eLA

topics. These PC members contributed timely and thoughtful paper reviews and were crucial to the workshop success. We plan to re-invite many (if not all) of them this year, including:

Ben Shneiderman (University of Maryland), Hal Daume III (University of Maryland), Michael Bernstein (Stanford University), Krzysztof Gajos (Harvard University), Elena Glassman (Harvard University), Maria De-Arteaga (UT Austin), Alon Jacovi (Bar Ilan University), Matthew Kay (Northwestern University), Michael Terry (Google Research), Fan Du (Adobe Research), Victor Dibia (Microsoft Research), Vera Liao (Microsoft Research), Jenn Wortman Vaughan (Microsoft Research), Tim Miller (University of Melbourne), Zahra Ashkourt (IBM Research), Jim Chen (University of Washington), Erin Chiou (ASU Adapt Lab), Ian Covert (University of Washington), Shi Feng (University of Maryland), Ella Glikson (Bar Ilan University), Maia Jacobs (Northwestern University), Joseph Janizek (University of Washington), Retno Larasati (The Open University), Brian Lim (National University of Singapore), Hussein Mozannar (MIT), Ishan Nigam (UT Austin), Marissa Radensky (University of Washington), Gonzalo Ramos (Microsoft Research), Jakob Schoeffer (Karlsruhe Institute of Technology), Tom Williams (Colorado School of Mines), Ming Yin (Purdue University), Tony Zhang (Fortiss), and Joyce Zhou (Cornell University).

Participants interested in giving a presentation at the workshop will need to submit a short paper (2-6 pages). Submission types will include, but are not limited to, position papers summarizing authors' existing research in the area and how it relates to the workshop theme, papers that review the related literature and offer a new perspective, and papers that describe work-in-progress research projects. We will encourage submissions that present diverse viewpoints on the workshop topics, and encourage participation across relevant fields, such as AI, HCI, and cognitive psychology. We will also create and advertise an industry track that encourages practitioners (who don't usually come to CHI) to submit papers that offer an industrial perspective or real-world approach to the workshop theme. We will use EasyChair to collect and review these submissions. Each submission will be reviewed by at least two PC members and one organizing committee member. To accommodate as many participants as possible, we plan to maintain a 50%-60% acceptance rate (similar to last year), such that paper authors can get a chance to join our workshop. Among them, we plan to include 10-15 spotlight that will be presented for during paper sessions. The remaining papers will be presented as posters (onsite) and as 1-minute pre-recorded presentations (online) during coffee breaks (see the next section for details on the workshop schedule.)

As workshop organizers, we value and are committed to diversity, equity, and inclusion. We welcome and encourage the participation of people who identify with any historically marginalized or under-represented group. Further, the listed platforms and technologies should not be a barrier to the participation of anyone interested in this workshop. If the technologies listed do not accommodate participants' needs, we will work with participants to find alternative solutions.

5 WORKSHOP STRUCTURE

Based on our workshop at CHI 2022 and the interest shown by colleagues, we expect between 60 and 80 participants. As such, the

workshop is currently designed for 80 participants. If interested participants exceed this number after the initial advertisement of the workshop, we may adjust the workshop structure to accommodate up to 100 participants. We plan to organize our proposal as a single-day workshop, from 9:00 AM to 5:00 PM local time (including breaks), in a hybrid format. We hope most participants join the in-person workshop but will also plan for synchronous online access to the workshop. We intend to use a combination of Zoom (for synchronized talks) and Slack (for virtual and asynchronous question answering and online discussions). In our post-workshop survey, we found that participants were particularly impressed by the Slack space setup, which we intend to keep: we will have channels for workshop session, and threads for each accepted paper so participants can submit their targeted questions. We will strongly encourage keynote speakers, paper authors and panelists to actively monitor and respond accordingly. The organizers will also work with the technical team at CHI 2023 to utilize provided streaming methods (with captioning for accessibility), so as to minimize the jump across platforms.

The tentative workshop schedule is detailed in Table 1. Since one of our goals is to synthesize knowledge and expertise from various communities and spur impactful future research, the workshop will dedicate sufficient time for group discussions and activities (afternoon session) in addition to a knowledge-sharing and discussion in the form of a keynote, paper presentations, and an expert panel discussion (morning session). This will help connect participants that share similar interests and provide them with a chance to contribute and learn.

The morning session will begin with a keynote by Carrie Cai and Michael Terry from Google Research. Their immense experience working with medical experts, music composers, designers, large language model developers will provide rich insights into practitioners' needs, and spark interest and motivation in later workshop events. Participants will have the opportunity to share their accepted work with either paper or poster presentations. The morning session will include two paper sessions for spotlight authors to share their accepted work. The presentations will consist of 5-7 minutes lightning talks, followed by a joint (around 10 minute) Q&A session. These talks will be pre-recorded; Non-spotlight acceptances will also have pre-recorded videos at 1-3 minute lengths. We will post all these recordings online for asynchronous access, since 2022 participants found it overwhelming to play all videos during paper sessions. Concurrent with the planned coffee breaks (in the morning and afternoon sessions), we also plan to hold poster presentations to support more interactions between authors and participants, and play videos of the 1-3 minute presentations.

We will also host a discussion panel of experts with balanced perspectives from academia and industry, to form the discussion around our diverse research interests—trust calibration, human-AI teaming, understanding AI uncertainty, etc. We have sent an initial round of invitation to five experts and already have commitments from some of them: Steph Ballard (a Sr. Practice Development Manager at the Microsoft Office of Responsible AI), Chris Albon (Director of Machine Learning at the Wikimedia Foundation with domain expertise in content moderation), Aaron Halfaker (Principal Researcher, Microsoft; Creator of open infrastructure for ML in Wikipedia), Curtiz Langlotz (Professor of Radiology and Biomedical

Slot	Theme
09:00 – 09:15 (15min)	Welcome
09:15 – 10:15 (65min)	Keynote talk, potentially by Carrie Cai / Michael Terry (Google PAIR)
10:15 – 10:45 (30min)	Paper sessions 1
10:15 – 10:45 (30min)	Coffee break (concurrent with poster presentations)
10:45 – 11:30 (45min)	Panel with experts that have diverse and well-balanced expertise
11:30 – 12:00 (30min)	Paper sessions 2
12:00 – 13:00 (60min)	Lunch break
13:00 – 14:30 (90min)	Group activity 1 (<i>60 min discussion + 30 min group result sharing</i>)
14:30 – 15:00 (30min)	Coffee break (concurrent with poster presentations)
15:00 – 16:30 (90min)	Group activity 2 (<i>60 min discussion + 30 min group result sharing</i>)
16:30 – 16:45 (15min)	Closing remarks

Table 1: Tentative schedule for the proposed single day workshop. The workshop will dedicate sufficient time for group discussions and activities (afternoon session) in addition to a knowledge-sharing and discussion in the form of a keynote, paper presentations, and an expert panel discussion (morning session).

Informatics at Stanford working on AI in Healthcare domains), and (4) Heather Reed (a Data Scientist at ASAPP³ AI+Customer Service agent teams, studying users’ trust on AI augmentation features in ASAPP product).

In the afternoon, we plan to allocate adequate time for two sessions of in-depth group activities, each session containing an one-hour within-group discussion, and a half-hour between-grouping insight sharing. The groups will be in the form of “birds-of-feather” discussions and practices around several topics, including measures, challenges, and mechanisms and interactions for shaping trust. We will finalize the group activities based on the number of participants and their interests, but some initial ideas include,

- (1) a *debating format*, where two groups are paired to represent the claims and counterclaims relating to themes within human-AI trust, so to motivate people to play devil’s advocates to each others ideas. The organizers would provide inspirational questions, as well as imaginary use scenarios that can ground these discussion (e.g., in high-stake domains like education, medical, etc.)
- (2) *concept mapping* around definitions, measures, and factors for appropriate trust and reliance. Groups’ could collaborate to enumerate and discuss relevant concepts—predictability, uncertainty, trust, reliance, adaptability, etc. and identify their overlaps and relationships.
- (3) *ideation* for solutions for shaping trust for particular use cases, such as news recommender systems or autonomous vehicles; here groups can brainstorm possible solutions, including system interactions, explanations, or visualizations, and iterate on these ideas with input from other groups. Outputs of this activity might consist of a set of solution ideas or low-fidelity mockups for system designs.
- (4) *on-the-spot paper writing and reviewing*: where participants come up with one research idea, or an imaginary paper they would like to write around the topic of trust and reliance. Groups’ output might be an abstract, certain teaser figures

illustrating the core idea, or compelling use cases. Then, participants will review these deliveries, hopefully to help better articulate what aspects people would care about around a particular research idea.

Each group will be moderated by at least one organizer; we will also encourage paper authors to join groups related to their paper topics and share their posters within the group, so they can have more dedicated discussions around the broader theme, but in the context of their own work. For hybrid participation, activities will make use of collaborative virtual environments like Google Documents and Miro boards.

We will host the paper lightening talks, posters, and group sharings in Google Slides and on the website, and we will later convert them into medium posts to share with the broader audience.

6 POST-WORKSHOP PLAN

We are now in the process of synthesizing our findings from the CHI 2022 workshop as a medium post and potentially as a journal submission. Similarly for CHI 2023, in order to reach a larger audience, we will upload the recorded sessions on YouTube, publish group activity outcomes as Medium posts, and share all the materials on social media.

7 CALL FOR PARTICIPATION: WORKSHOP ON TRUST AND RELIANCE IN AI-HUMAN TEAMS (TRAIT)

As humans increasingly interact with AI systems during decision-making, creative tasks, and other workflows, appropriate trust and reliance are necessary to ensure proper usage and adoption of these systems. For example, people should understand when to trust or rely on an algorithm’s outputs and when to override them. While significant research focus has aimed to measure and promote trust in human-AI interaction, the field lacks synthesized results across contexts, formalized key concepts, and definitions. The workshop will provide a venue to explore three broad aspects related to human-AI trust: (1) How do we clarify definitions and frameworks

³<https://www.asapp.com/>

relevant to human-AI trust and reliance (e.g., what does trust mean in different contexts)? (2) How do we measure trust and reliance? And, (3) How do we shape trust and reliance?

Themes include, but are not limited to:

- Definitions of trust and reliance.
- Human-human trust and lessons from social sciences.
- Qualitative (e.g., user reflection) and quantitative methods (e.g., usage, adoption, team performance, etc.) for evaluating trust and reliance.
- Tradeoffs with other objectives (e.g., team performance, creativity, etc)
- Solutions (and their limitations) for promoting appropriate trust (e.g., XAI, control mechanisms, human agency, communicating uncertainty etc).
- Safety mechanisms for when trust is broken.

We welcome submissions to either the **research** track, or the **industry** track. Research track papers may present a new position, summarize existing research, or discuss in-progress works. Industrial track papers should focus more on providing industry perspective, e.g., How do practitioners in startups, non-profits, etc. think about trust and reliance on AI models? How do they view the theoretical work and case studies coming out of academia?

The submission should use ACM single column format, and should take 4-10 pages. At least one author must register and attend the workshop. Submission will be reviewed by program committee and accepted papers will be posted on the workshop website and shared via social media. The workshop will feature keynote by Carrie Cai and Michael Terry (Google Research); moderated panel of diverse experts; short talks or posters by authors; and small group activities to explore workshop themes and promote collaborations.

Important Dates:

- Submission: February 23, 2023 (EasyChair)
- Notifications: March 14, 2023
- Camera Ready: April 11, 2023
- Workshop: TBD (between April 23-28, 2023)

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